

Perimeter And Area

Exercise 11.1

Question 1:

The length and the breadth of a rectangular piece of land are 500 m and 300 m respectively. Find

(i) its area

(ii) the cost of the land, if 1 m² of the land costs Rs 10,000.

Answer:

(i) Area = Length × Breadth

$$= 500 \times 300$$

$$= 150000 \text{ m}^2$$

(ii) Cost of 1 m² land = Rs 10000

$$\text{Cost of } 150000 \text{ m}^2 \text{ land} = 10000 \times 150000 = \text{Rs } 1500000000$$

Question 2:

Find the area of a square park whose perimeter is 320 m.

Answer:

$$\text{Perimeter} = 320 \text{ m}$$

$$4 \times \text{Length of the side of park} = 320$$

$$\text{Length of the side of park} = \frac{320}{4} = 80 \text{ m}$$

$$\text{Area} = (\text{Length of the side of park})^2 = (80)^2 = 6400 \text{ m}^2$$

Question 3:

Find the breadth of a rectangular plot of land, if its area is 440 m^2 and the length is 22 m. Also find its perimeter.

Answer:

$$\text{Area} = \text{Length} \times \text{Breadth} = 440 \text{ m}^2$$

$$22 \times \text{Breadth} = 440$$

$$\text{Breadth} = \frac{440}{22} = 20 \text{ m}$$

$$\text{Perimeter} = 2 (\text{Length} + \text{Breadth})$$

$$= 2 (22 + 20) = 2 (42) = 84 \text{ m}^2$$

Question 4:

The perimeter of a rectangular sheet is 100 cm. If the length is 35 cm, find its breadth. Also find the area.

Answer:

$$\text{Perimeter} = 2 (\text{Length} + \text{Breadth}) = 100 \text{ cm}$$

$$2 (35 + \text{Breadth}) = 100$$

$$35 + B = 50$$

$$B = 50 - 35 = 15 \text{ cm}$$

$$\text{Area} = \text{Length} \times \text{Breadth} = 35 \times 15 = 525 \text{ cm}^2$$

Question 5:

The area of a square park is the same as of a rectangular park. If the side of the square park is 60 m and the length of the rectangular park is 90 m, find the breadth of the rectangular park.

Answer:

$$\text{Area of square park} = (\text{One of its sides})^2 = (60)^2 = 3600 \text{ m}^2$$

$$\text{Area of rectangular park} = \text{Length} \times \text{Breadth} = 3600$$

$$90 \times \text{Breadth} = 3600$$

$$\text{Breadth} = 40 \text{ m}$$

Question 6:

A wire is in the shape of a rectangle. Its length is 40 cm and breadth is 22 cm. If the same wire is rebent in the shape of a square, what will be the measure of each side. Also find which shape encloses more area?

Answer:

$$\text{Perimeter of rectangle} = \text{Perimeter of square}$$

$$2 (\text{Length} + \text{Breadth}) = 4 \times \text{Side}$$

$$2 (40 + 22) = 4 \times \text{Side}$$

$$2 \times 62 = 4 \times \text{Side}$$

$$\text{Side} = \frac{124}{4} = 31 \text{ cm}$$

$$\text{Area of rectangle} = 40 \times 22 = 880 \text{ cm}^2$$

$$\text{Area of square} = (\text{Side})^2 = 31 \times 31 = 961 \text{ cm}^2$$

Therefore, the square-shaped wire encloses more area.

Question 7:

The perimeter of a rectangle is 130 cm. If the breadth of the rectangle is 30 cm, find its length. Also find the area of the rectangle.

Answer:

$$\text{Perimeter} = 2 (\text{Length} + \text{Breadth}) = 130$$

$$2 (\text{Length} + 30) = 130$$

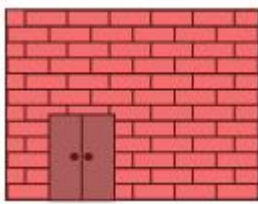
$$\text{Length} + 30 = 65$$

$$\text{Length} = 65 - 30 = 35 \text{ cm}$$

$$\text{Area} = \text{Length} \times \text{Breadth} = 35 \times 30 = 1050 \text{ cm}^2$$

Question 8:

A door of length 2 m and breadth 1 m is fitted in a wall. The length of the wall is 4.5 m and the breadth is 3.6 m (see the given figure). Find the cost of white washing the wall, if the rate of white washing the wall is Rs 20 per m^2 .



Answer:

$$\text{Area of wall} = 4.5 \times 3.6 = 16.2 \text{ m}^2$$

$$\text{Area of door} = 2 \times 1 = 2 \text{ m}^2$$

$$\text{Area to be white-washed} = 16.2 - 2 = 14.2 \text{ m}^2$$

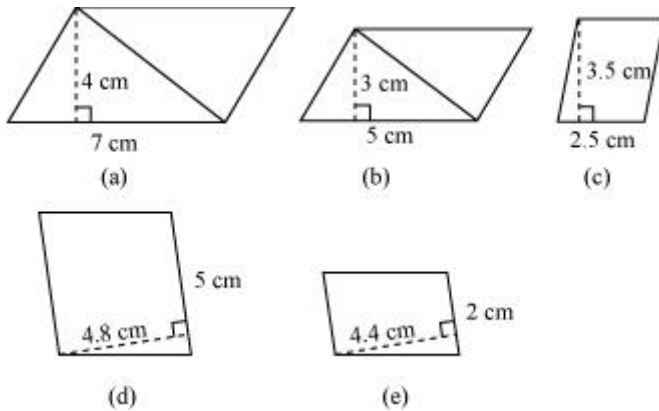
$$\text{Cost of white-washing 1 m}^2 \text{ area} = \text{Rs } 20$$

$$\therefore \text{Cost of white-washing 14.2 m}^2 \text{ area} = 14.2 \times 20 = \text{Rs } 284$$

Exercise 11.2

Question 1:

Find the area of each of the following parallelograms:



Answer:

Area of parallelogram = Base $\times$ Height

(a) Height= 4 cm

Base = 7 cm

Area of parallelogram = $7 \times 4 = 28 \text{ cm}^2$

(b) Height= 3 cm

Base = 5 cm

Area of parallelogram = $5 \times 3 = 15 \text{ cm}^2$

(c) Height= 3.5 cm

Base = 2.5 cm

Area of parallelogram = $2.5 \times 3.5 = 8.75 \text{ cm}^2$

(d) Height= 4.8 cm

Base = 5 cm

Area of parallelogram = $5 \times 4.8 = 24 \text{ cm}^2$

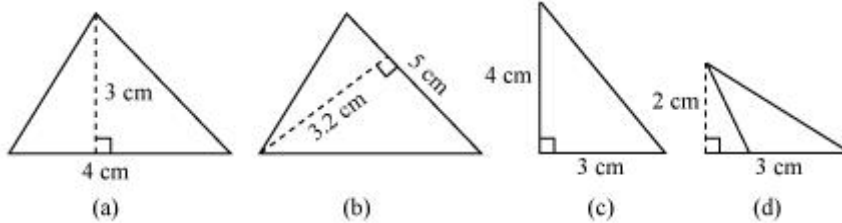
(e) Height= 4.4 cm

Base = 2 cm

Area of parallelogram = $2 \times 4.4 = 8.8 \text{ cm}^2$

Question 2:

Find the area of each of the following triangles:



Answer:

$$\text{Area of triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

(a) Base = 4 cm, height = 3 cm

$$\text{Area} = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$$

(b) Base = 5 cm, height = 3.2 cm

$$\text{Area} = \frac{1}{2} \times 5 \times 3.2 = 8 \text{ cm}^2$$

(c) Base = 4 cm, height = 3 cm

$$\text{Area} = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$$

(d) Base = 3 cm, height = 2 cm

$$\text{Area} = \frac{1}{2} \times 2 \times 3 = 3 \text{ cm}^2$$

Question 3:

Find the missing values:

So No	Base	Height	Area of parallelogram
a.	20 cm	-	246 cm ²
b.	-	15 cm	154.5 cm ²
c.	-	8.4 cm	48.72 cm ²
d.	15.6 cm	-	16.38 cm ²

Answer:

Area of parallelogram = Base × Height

(a) $b = 20$ cm

$h = ?$

Area = 246 cm²

$20 \times h = 246$

$$h = \frac{246}{20} = 12.3 \text{ cm}$$

Therefore, the height of such parallelogram is 12.3 cm.

(b) $b = ?$

$h = 15$ cm

Area = 154.5 cm²

$b \times 15 = 154.5$

$b = 10.3$ cm

Therefore, the base of such parallelogram is 10.3 cm.

(c) $b = ?$

$$h = 8.4 \text{ cm}$$

$$\text{Area} = 48.72 \text{ cm}^2$$

$$b \times 8.4 = 48.72$$

$$b = \frac{48.72}{8.4} = 5.8 \text{ cm}$$

Therefore, the base of such parallelogram is 5.8 cm.

(d) $b = 15.6 \text{ cm}$

$$h = ?$$

$$\text{Area} = 16.38 \text{ cm}^2$$

$$15.6 \times h = 16.38$$

$$h = \frac{16.38}{15.6} = 1.05 \text{ cm}$$

Therefore, the height of such parallelogram is 1.05 cm.

Question 4:

Find the missing values:

Base	Height	Area of triangle
15 cm	_____	87 cm ²
_____	31.4 mm	1256 mm ²

22 cm	_____	170.5 cm ²
-------	-------	-----------------------

Answer:

$$\text{Area of triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

(a) $b = 15 \text{ cm}$

$h = ?$

$$\text{Area} = \frac{1}{2} \times b \times h = 87 \text{ cm}^2$$

$$\frac{1}{2} \times 15 \times h = 87 \text{ cm}^2$$

$$h = \frac{87 \times 2}{15} = 11.6 \text{ cm}$$

Therefore, the height of such triangle is 11.6 cm.

(b) $b = ?$

$h = 31.4 \text{ mm}$

$$\text{Area} = \frac{1}{2} \times b \times h = 1256 \text{ mm}^2$$

$$\frac{1}{2} \times b \times 31.4 = 1256$$

$$b = \frac{1256 \times 2}{31.4} = 80 \text{ mm}$$

Therefore, the base of such triangle is 80 mm.

(c) $b = 22 \text{ cm}$

$h = ?$

$$\text{Area} = \frac{1}{2} \times b \times h = 170.5 \text{ cm}^2$$

$$\frac{1}{2} \times 22 \times h = 170.5 \text{ cm}^2$$

$$h = \frac{170.5 \times 2}{22} = 15.5 \text{ cm}$$

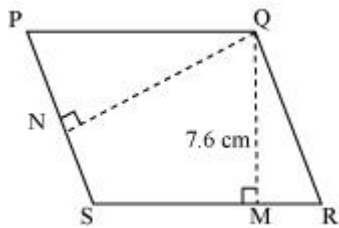
Therefore, the height of such triangle is 15.5 cm.

Question 5:

PQRS is a parallelogram (see the given figure). QM is the height from Q to SR and QN is the height from Q to PS. If SR = 12 cm and QM = 7.6 cm. Find:

(a) the area of the parallelogram PQRS

(b) QN, if PS = 8 cm



Answer:

(a) Area of parallelogram = Base $\times$ Height = SR $\times$ QM

$$= 7.6 \times 12 = 91.2 \text{ cm}^2$$

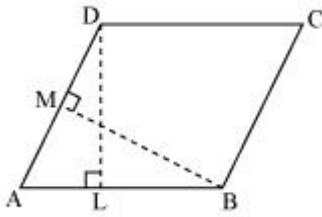
(b) Area of parallelogram = Base $\times$ Height = PS $\times$ QN = 91.2 cm²

$$\text{QN} \times 8 = 91.2$$

$$\text{QN} = \frac{91.2}{8} = 11.4 \text{ cm}$$

Question 6:

DL and BM are the heights on sides AB and AD respectively of parallelogram ABCD (see the given figure). If the area of the parallelogram is 1470 cm^2 , $AB = 35 \text{ cm}$ and $AD = 49 \text{ cm}$, find the length of BM and DL.



Answer:

Area of parallelogram = Base $\times$ Height = $AB \times DL$

$$1470 = 35 \times DL$$

$$DL = \frac{1470}{35} = 42 \text{ cm}$$

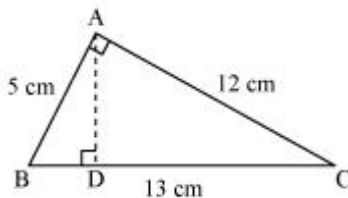
Also, $1470 = AD \times BM$

$$1470 = 49 \times BM$$

$$BM = \frac{1470}{49} = 30 \text{ cm}$$

Question 7:

$\triangle ABC$ is right angled at A (see the given figure). AD is perpendicular to BC. If $AB = 5 \text{ cm}$, $BC = 13 \text{ cm}$ and $AC = 12 \text{ cm}$, Find the area of $\triangle ABC$. Also find the length of AD.



Answer:

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times 5 \times 12 = 30 \text{ cm}^2$$

$$\text{Also, area of triangle} = \frac{1}{2} \times \text{AD} \times \text{BC}$$

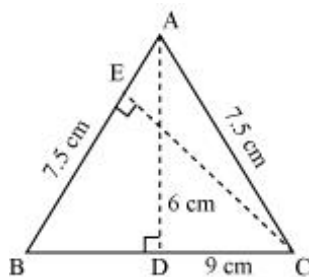
$$30 = \frac{1}{2} \times \text{AD} \times 13$$

$$\frac{30 \times 2}{13} = \text{AD}$$

$$\text{AD} = 4.6 \text{ cm}$$

Question 8:

ΔABC is isosceles with $AB = AC = 7.5 \text{ cm}$ and $BC = 9 \text{ cm}$ (see the given figure). The height AD from A to BC , is 6 cm . Find the area of ΔABC . What will be the height from C to AB i.e., CE ?



Answer:

$$\begin{aligned} \text{Area of } \Delta ABC &= \frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times \text{BC} \times \text{AD} \\ &= \frac{1}{2} \times 9 \times 6 = 27 \text{ cm}^2 \end{aligned}$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times \text{AB} \times \text{CE}$$

$$27 = \frac{1}{2} \times 7.5 \times \text{CE}$$

$$\text{CE} = 7.2 \text{ cm}$$

Exercise 11.3

Question 1:

Find the circumference of the circles with the following radius: (Take $\pi = \frac{22}{7}$)

(a) 14 cm (b) 28 mm (c) 21 cm

Answer:

(a) $r = 14$ cm

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 14 = 88 \text{ cm}$$

(b) $r = 28$ mm

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 28 = 176 \text{ mm}$$

(c) $r = 21$ cm

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 21 = 132 \text{ cm}$$

Question 2:

Find the area of the following circles, given that:

(a) radius = 14 mm (Take $\pi = \frac{22}{7}$) (b) diameter = 49 m

(c) radius = 5 cm

Answer:

(a) $r = 14$ mm

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ mm}^2$$

(b) $d = 49 \text{ m}$

$$r = \frac{49}{2} \text{ m}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times \frac{49}{2} \times \frac{49}{2} = 1886.5 \text{ m}^2$$

(c) $r = 5 \text{ cm}$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 5 \times 5 = \frac{550}{7} = 78.57 \text{ cm}^2$$

Question 3:

If the circumference of a circular sheet is 154m, find its radius. Also find the area of

the sheet. (Take $\pi = \frac{22}{7}$)

Answer:

$$\text{Circumference} = 2\pi r = 154 \text{ m}$$

$$2 \times \frac{22}{7} \times r = 154$$

$$r = 154 \times \frac{7}{44} = \frac{49}{2} = 24.5 \text{ m}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times r^2$$

$$= \frac{22}{7} \times \frac{49}{2} \times \frac{49}{2} = 1886.5 \text{ m}^2$$

Question 4:

A gardener wants to fence a circular garden of diameter 21 m. Find the length of the rope he needs to purchase, if he makes 2 rounds of fence. Also find the

costs of the rope, if it cost Rs 4 per meter. (Take $\pi = \frac{22}{7}$)

Answer:

$$d = 21 \text{ m}$$

$$r = \frac{21}{2} \text{ m}$$

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times \frac{21}{2} = 66 \text{ m}$$

$$\text{Length of rope required for fencing} = 2 \times 66 \text{ m} = 132 \text{ m}$$

$$\text{Cost of 1 m rope} = \text{Rs } 4$$

$$\text{Cost of 132 m rope} = 4 \times 132 = \text{Rs } 528$$

Question 5:

From a circular sheet of radius 4 cm, a circle of radius 3 cm is removed. Find the area of the remaining sheet. (Take $\pi = 3.14$)



Answer:

$$\text{Outer radius of circular sheet} = 4 \text{ cm}$$

$$\text{Inner radius of circular sheet} = 3 \text{ cm}$$

$$\text{Remaining area} = 3.14 \times 4 \times 4 - 3.14 \times 3 \times 3$$

$$= 50.24 - 28.26$$

$$= 21.98 \text{ cm}^2$$

Question 6:

Saima wants to put a lace on the edge of a circular table cover of diameter 1.5 m. Find the length of the lace required and also find its cost if one meter of the lace costs Rs 15.(Take $\pi = 3.14$)

Answer:

$$\text{Circumference} = 2\pi r$$

$$= 2 \times 3.14 \times \frac{d}{2}$$

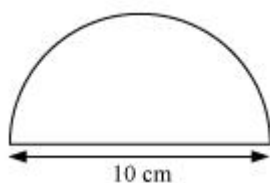
$$= 2 \times 3.14 \times \frac{1.5}{2} = 4.71 \text{ m}$$

$$\text{Cost of 1 m lace} = \text{Rs } 15$$

$$\text{Cost of 4.71 m lace} = 4.71 \times 15 = \text{Rs } 70.65$$

Question 7:

Find the perimeter of the adjoining figure, which is a semicircle including its diameter.



Answer:

$$\text{Radius} = 5 \text{ cm}$$

$$\text{Length of curved part} = \pi r$$

$$= \frac{22}{7} \times 5$$

$$= 15.71 \text{ cm}$$

Total perimeter = Length of curved part + Length of diameter

$$= 15.71 + 10 = 25.71 \text{ cm}$$

Question 8:

Find the cost of polishing a circular table-top of diameter 1.6 m, if the rate of polishing is Rs 15/m² . (Take $\pi = 3.14$)

Answer:

Diameter = 1.6 m

$$\text{Radius} = \frac{1.6}{2} = 0.8 \text{ m}$$

$$\text{Area} = 3.14 \times 0.8 \times 0.8$$

$$= 2.0096 \text{ m}^2$$

Cost for polishing 1 m² area = Rs 15

$$\text{Cost for polishing } 2.0096 \text{ m}^2 \text{ area} = 15 \times 2.0096 = 30.14$$

Therefore, it will cost Rs 30.14 for polishing such circular table.

Question 9:

Shazli took a wire of length 44 cm and bent it into the shape of a circle. Find the radius of that circle. Also find its area. If the same wire is bent into the shape of a square, what will be the length of each of its sides? Which figure

encloses more area, the circle or the square? (Take $\pi = \frac{22}{7}$)

Answer:

$$\text{Circumference} = 2\pi r = 44 \text{ cm}$$

$$2 \times \frac{22}{7} \times r = 44$$

$$r = 7 \text{ cm}$$

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

If the wire is bent into a square, then the length of each side would be $= \frac{44}{4} = 11 \text{ cm}$

$$\text{Area of square} = (11)^2 = 121 \text{ cm}^2$$

Therefore, circle encloses more area.

Question 10:

From a circular card sheet of radius 14 cm, two circles of radius 3.5 cm and a rectangle of length 3 cm and breadth 1 cm are removed (as shown in the

following figure). Find the area of the remaining sheet. (Take $\pi = \frac{22}{7}$)



Answer:

$$\text{Area of bigger circle} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$$

$$\text{Area of 2 small circles} = 2 \times \pi r^2 = 2 \times \frac{22}{7} \times 3.5 \times 3.5 = 77 \text{ cm}^2$$

$$\text{Area of rectangle} = \text{Length} \times \text{Breadth} = 3 \times 1 = 3 \text{ cm}^2$$

$$\text{Remaining area of sheet} = 616 - 77 - 3 = 536 \text{ cm}^2$$

Question 11:

A circle of radius 2 cm is cut out from a square piece of an aluminium sheet of side 6 cm. What is the area of the left over aluminium sheet? (Take $\pi = 3.14$)

Answer:

$$\text{Area of square-shaped sheet} = (\text{Side})^2 = (6)^2 = 36 \text{ cm}^2$$

$$\text{Area of circle} = 3.14 \times 2 \times 2 = 12.56 \text{ cm}^2$$

$$\text{Remaining area of sheet} = 36 - 12.56 = 23.44 \text{ cm}^2$$

Question 12:

The circumference of a circle is 31.4 cm. Find the radius and the area of the circle? (Take $\pi = 3.14$)

Answer:

$$\text{Circumference} = 2\pi r = 31.4 \text{ cm}$$

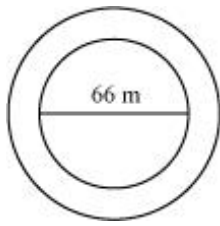
$$2 \times 3.14 \times r = 31.4$$

$$r = 5 \text{ cm}$$

$$\text{Area} = 3.14 \times 5 \times 5 = 78.50 \text{ cm}^2$$

Question 13:

A circular flower bed is surrounded by a path 4 m wide. The diameter of the flower bed is 66 m. What is the area of this path? ($\pi = 3.14$)



Answer:

$$\text{Radius of flower bed} = \frac{66}{2} = 33 \text{ m}$$

$$\text{Radius of flower bed and path together} = 33 + 4 = 37 \text{ m}$$

$$\text{Area of flower bed and path together} = 3.14 \times 37 \times 37 = 4298.66 \text{ m}^2$$

$$\text{Area of flower bed} = 3.14 \times 33 \times 33 = 3419.46 \text{ m}^2$$

$$\begin{aligned} \text{Area of path} &= \text{Area of flower bed and path together} - \text{Area of flower bed} \\ &= 4298.66 - 3419.46 = 879.20 \text{ m}^2 \end{aligned}$$

Question 14:

A circular flower garden has an area of 314 m^2 . A sprinkler at the centre of the garden can cover an area that has a radius of 12 m. Will the sprinkler water the entire garden? (Take $\pi = 3.14$)

Answer:

$$\text{Area} = \pi r^2 = 314 \text{ m}^2$$

$$3.14 \times r^2 = 314$$

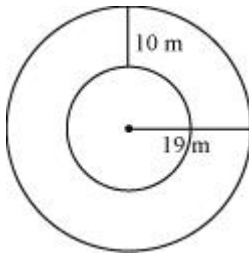
$$r^2 = 100$$

$$r = 10 \text{ m}$$

Yes, the sprinkler will water the whole garden.

Question 15:

Find the circumference of the inner and the outer circles, shown in the adjoining figure? (Take $\pi = 3.14$)



Answer:

Radius of outer circle = 19 m

Circumference = $2\pi r = 2 \times 3.14 \times 19 = 119.32$ m

Radius of inner circle = $19 - 10 = 9$ m

Circumference = $2\pi r = 2 \times 3.14 \times 9 = 56.52$ m

Question 16:

How many times a wheel of radius 28 cm must rotate to go 352 m?

(Take $\pi = \frac{22}{7}$)

Answer:

$r = 28$ cm

Circumference = $2\pi r = 2 \times \frac{22}{7} \times 28 = 176$ cm

Number of rotations = $\frac{\text{Total distance to be covered}}{\text{Circumference of wheel}} = \frac{352\text{m}}{176\text{cm}} = \frac{35200}{176} = 200$

Therefore, it will rotate 200 times.

Question 17:

The minute hand of a circular clock is 15 cm long. How far does the tip of the minute hand move in 1 hour. (Take $\pi = 3.14$)

Answer:

Distance travelled by the tip of minute hand = Circumference of the clock

$$= 2\pi r = 2 \times 3.14 \times 15$$

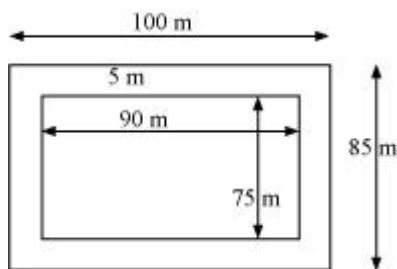
$$= 94.2 \text{ cm}$$

Exercise 11.4

Question 1:

A garden is 90 m long and 75 m broad. A path 5 m wide is to be built outside and around it. Find the area of the path. Also find the area of the garden in hectare.

Answer:



Length (l) of garden = 90 m

Breadth (b) of garden = 75 m

$$\text{Area of garden} = l \times b = 90 \times 75 = 6750 \text{ m}^2$$

From the figure, it can be observed that the new length and breadth of the garden, when path is also included, are 100m and 85m respectively.

$$\text{Area of the garden including the path} = 100 \times 85 = 8500 \text{ m}^2$$

$$\text{Area of path} = \text{Area of the garden including the path} - \text{Area of garden}$$

$$= 8500 - 6750 = 1750 \text{ m}^2$$

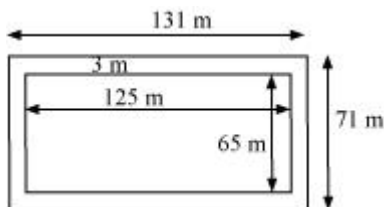
$$1 \text{ hectare} = 10000 \text{ m}^2$$

$$\text{Therefore, area of garden in hectare} = \frac{6750}{10000} = 0.675 \text{ hectare}$$

Question 2:

A 3 m wide path runs outside and around a rectangular park of length 125 m and breadth 65 m. Find the area of the path.

Answer:



$$\text{Length (l) of park} = 125 \text{ m}$$

$$\text{Breadth (b) of park} = 65 \text{ m}$$

$$\text{Area of park} = l \times b = 125 \times 65 = 8125 \text{ m}^2$$

From the figure, it can be observed that the new length and breadth of the park, when path is also included, are 131 m and 71 m respectively.

$$\text{Area of the park including the path} = 131 \times 71 = 9301 \text{ m}^2$$

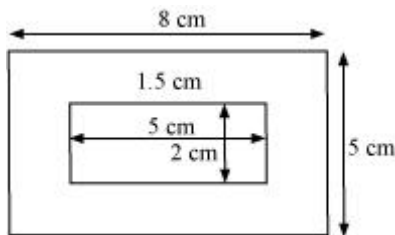
$$\text{Area of path} = \text{Area of the park including the path} - \text{Area of park}$$

$$= 9301 - 8125 = 1176 \text{ m}^2$$

Question 3:

A picture is painted on a cardboard 8 cm long and 5 cm wide such that there is a margin of 1.5 cm along each of its sides. Find the total area of the margin.

Answer:



Length (l) of cardboard = 8 cm

Breadth (b) of cardboard = 5 cm

Area of cardboard including margin = $l \times b = 8 \times 5 = 40 \text{ cm}^2$

From the figure, it can be observed that the new length and breadth of the cardboard, when margin is not included, are 5 cm and 2 cm respectively.

Area of the cardboard not including the margin = $5 \times 2 = 10 \text{ cm}^2$

Area of the margin = Area of cardboard including the margin – Area of cardboard not including the margin

$= 40 - 10 = 30 \text{ cm}^2$

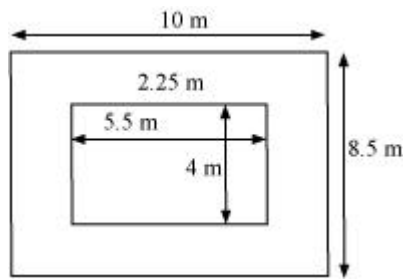
Question 4:

A verandah of width 2.25 m is constructed all along outside a room which is 5.5 m long and 4 m wide. Find:

(i) the area of the verandah

(ii) the cost of cementing the floor of the verandah at the rate of Rs 200 per m^2 .

Answer:



(i)

Length (l) of room = 5.5 m

Breadth (b) of room = 4 m

Area of room = $l \times b = 5.5 \times 4 = 22 \text{ m}^2$

From the figure, it can be observed that the new length and breadth of the room, when verandah is also included, are 10 m and 8.5 m respectively.

Area of the room including the verandah = $10 \times 8.5 = 85 \text{ m}^2$

Area of verandah = Area of the room including the verandah – Area of room

= $85 - 22 = 63 \text{ m}^2$

(ii)

Cost of cementing 1 m^2 area of the floor of the verandah = Rs 200

Cost of cementing 63 m^2 area of the floor of the verandah = 200×63

= Rs 12600

Question 5:

A path 1 m wide is built along the border and inside a square garden of side 30 m. Find:

(i) the area of the path

(ii) the cost of planting grass in the remaining portion of the garden at the rate of Rs 40 per m^2 .

Answer:

(i) Side of the square garden = 30 m and

Width of the path along the border = 1 m

Side of square garden without path = $30 - (1 + 1) = 30 - 2 = 28$ m

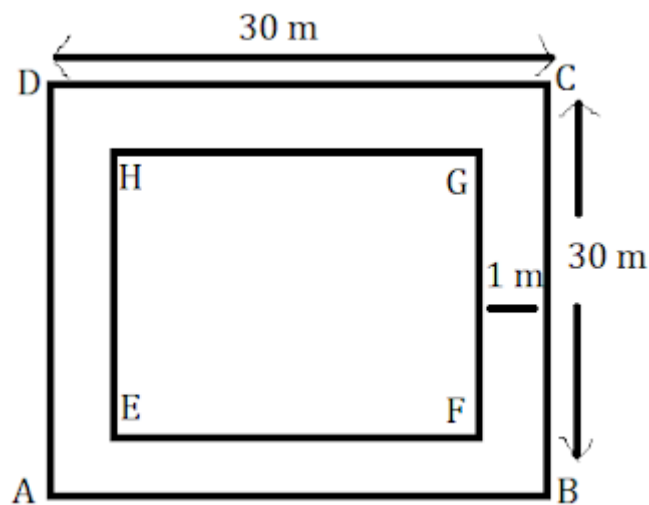
Now Area of path = Area of ABCD - Area of EFGH

$$= (AB \times AD) - (EF \times EH)$$

$$= (30 \times 30) - (28 \times 28)$$

$$= 900 - 784$$

$$= 116 \text{ m}^2$$



(ii) Area of remaining portion = $28 \times 28 = 784 \text{ m}^2$

The cost of planting grass in 1 m^2 of the garden = Rs 40

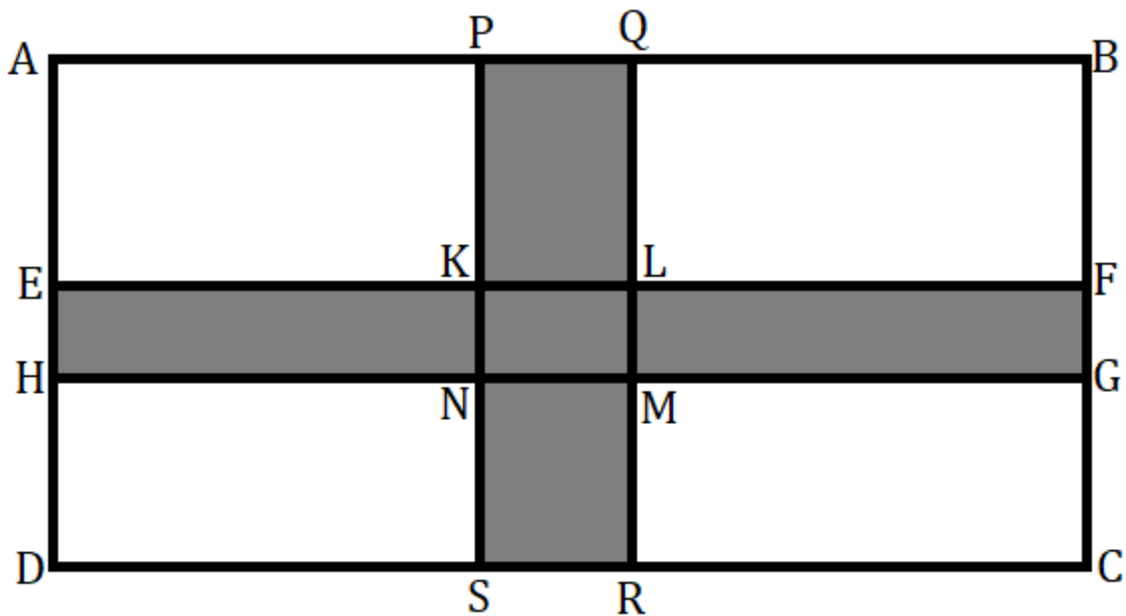
The cost of planting grass in 784 m^2 of the garden = $\text{Rs } 40 \times 784 = \text{Rs } 31,360$

Question 6:

Two cross roads, each of width 10 m, cut at right angles through the centre of a rectangular park of length 700 m and breadth 300 m and parallel to its sides. Find the area of the roads. Also find the area of the park excluding cross roads. Give the answer in hectares.

Answer:

Here, PQ = 10 m and PS = 300 m, EH = 10 m and EF = 700 m And KL = 10 m and KN = 10 m



Area of roads = Area of PQRS + Area of EFGH - Area of KLMN

[$\because$ KLMN is taken twice, which is to be subtracted] = $(PS \times PQ) + (EF \times EH) - (KL \times KN)$

$$= (300 \times 10) + (700 \times 10) - (10 \times 10)$$

$$= 3000 + 7000 - 100$$

$$= 9,900 \text{ m}^2$$

Area of road in hectares, $1\text{m}^2 = 1/10000$ hectares

$$\therefore 9,900 \text{ m}^2 = 9900/10000 = 0.99 \text{ hectares}$$

Now, Area of park excluding cross roads

= Area of park - Area of road

= $(AB \times AD) - 9,900 = (700 \times 300) - 9,900$

= $2,10,000 - 9,900$

= $2,00,100 \text{ m}^2$

= $200100/10000 \text{ hectares} = 20.01 \text{ hectares.}$

Question 7:

Through a rectangular field of length 90 m and breadth 60 m, two roads are constructed which are parallel to the sides and cut each other at right angles through the centre of the fields. If the width of each road is 3 m, find

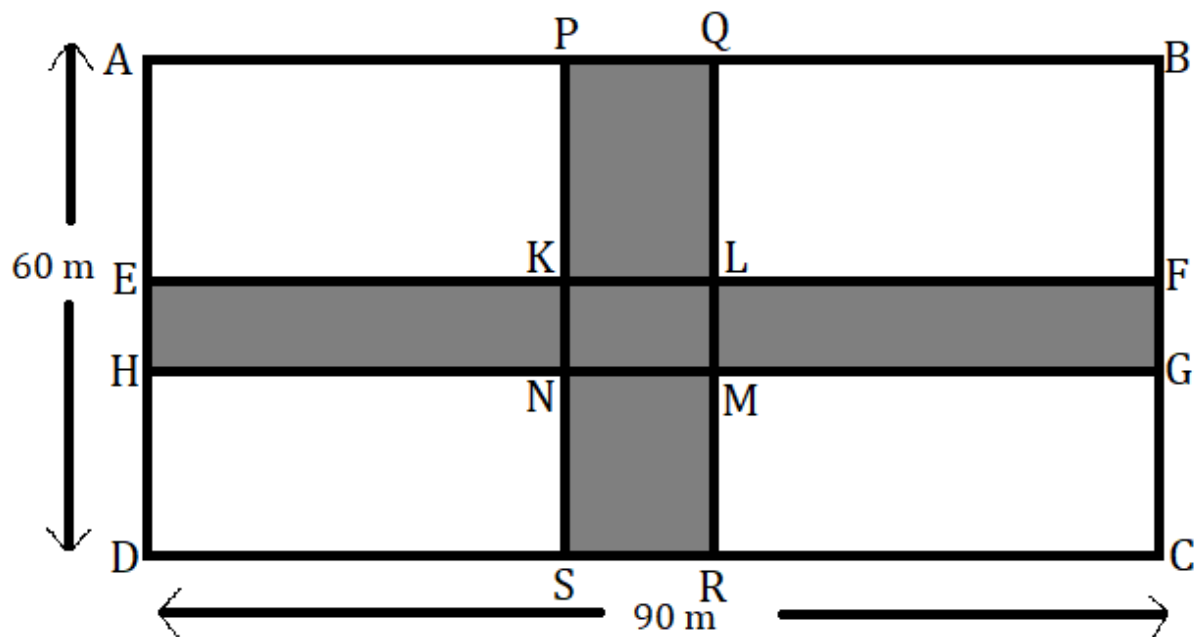
(i) the area covered by the roads.

(ii) the cost of constructing the roads at the rate of Rs 110 per m^2 .

Answer:

(i) Here, $PQ = 3 \text{ m}$ and $PS = 60 \text{ m}$, $EH = 3 \text{ m}$ and

$EF = 90 \text{ m}$ and $KL = 3 \text{ m}$ and $KN = 3 \text{ m}$



Area of roads = Area of PQRS + Area of EFGH - Area of KLMN

[$\because$ KLMN is taken twice, which is to be subtracted]

$$= (PS \times PQ) + (EF \times EH) - (KL \times KN)$$

$$= (60 \times 3) + (90 \times 3) - (3 \times 3)$$

$$= 180 + 270 - 9$$

$$= 441 \text{ m}^2$$

(ii) The cost of 1 m^2 constructing the roads = Rs 110

The cost of 441 m^2 constructing the roads = Rs 110 $\times$ 441 = Rs 48,510

Therefore, the cost of constructing the roads = Rs 48,510.

Question 8:

Pragya wrapped a cord around a circular pipe of radius 4 cm (adjoining figure) and cut off the length required of the cord. Then she wrapped it around a square box of side 4 cm (also shown). Did she have any cord left? ($\pi = 3.14$).



Answer:

Perimeter of the square = $4 \times \text{Side of the square} = 4 \times 4 = 16 \text{ cm}$

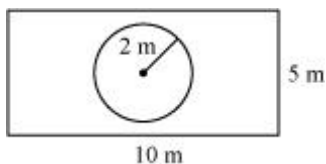
Perimeter of circular pipe = $2\pi r = 2 \times 3.14 \times 4 = 25.12 \text{ cm}$

Length of chord left with Pragya = $25.12 - 16 = 9.12 \text{ cm}$

Question 9:

The adjoining figure represents a rectangular lawn with a circular flower bed in the middle. Find:

- (i) the area of the whole land
- (ii) the area of the flower bed
- (iii) the area of the lawn excluding the area of the flower bed
- (iv) the circumference of the flower bed.



Answer:

Length of rectangular lawn = 10 m, breadth of the rectangular lawn = 5 m

And radius of the circular flower bed = 2 m

(i) Area of the whole land = length $\times$ breadth = $10 \times 5 = 50 \text{ m}^2$

(ii) Area of flower bed = πr^2

$$= 3.14 \times 2 \times 2 = 12.56 \text{ m}^2$$

(iii) Area of lawn excluding the area of the flower bed = area of lawn - area of flower bed

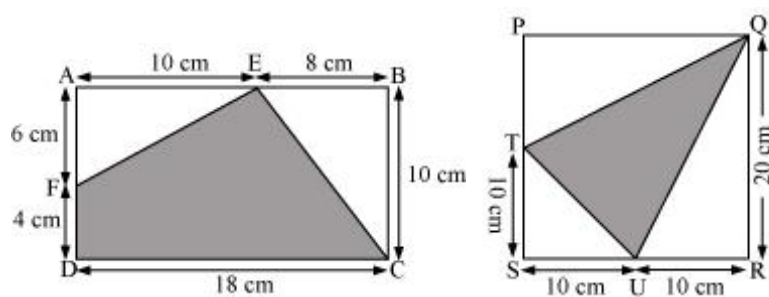
$$= 50 - 12.56$$

$$= 37.44 \text{ m}^2$$

(iv) The circumference of the flower bed = $2\pi r = 2 \times 3.14 \times 2 = 12.56 \text{ m}$.

Question 10:

In the following figures, find the area of the shaded portions:



Answer:

(i) Here, $AB = 18 \text{ cm}$, $BC = 10 \text{ cm}$, $AF = 6 \text{ cm}$, $AE = 10 \text{ cm}$ and $BE = 8 \text{ cm}$

Area of shaded portion = Area of rectangle ABCD - (Area of $\triangle FAE$ + area of $\triangle EBC$)

$$= (AB \times BC) - (1/2 \times AE \times AF + 1/2 \times BE \times BC)$$

$$= (18 \times 10) - (1/2 \times 10 \times 6 + 1/2 \times 8 \times 10)$$

$$= 180 - (30 + 40)$$

$$= 180 - 70$$

$$= 110 \text{ cm}^2.$$

(ii) Here, $SR = SU + UR = 10 + 10 = 20 \text{ cm}$, $QR = 20 \text{ cm}$

$$PQ = SR = 20 \text{ cm}$$

$$PT = PS - TS = (20 - 10) \text{ cm}$$

$TS = 10 \text{ cm}$, $SU = 10 \text{ cm}$, $QR = 20 \text{ cm}$ and $UR = 10 \text{ cm}$

Area of shaded region

$$= \text{Area of square PQRS} - \text{Area of } \triangle QPT - \text{Area of } \triangle TSU - \text{Area of } \triangle UQR$$

$$= (SR \times QR) - 1/2 \times PQ \times PT - 1/2 \times ST \times SU - 1/2$$

$$= 20 \times 20 - 1/2 \times 20 \times 10 - 1/2 \times 10 \times 10 - 1/2 \times 20 \times 10$$

$$= 400 - 100 - 50 - 100$$

$$= 150 \text{ cm}^2$$

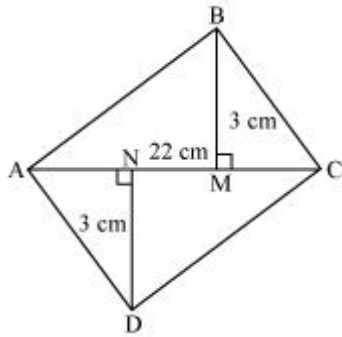
Question 11:

Find the area of the quadrilateral ABCD.

Here, $AC = 22 \text{ cm}$, $BM = 3 \text{ cm}$,

$DN = 3 \text{ cm}$, and

$BM \perp AC$, $DN \perp AC$



Answer:

Here, $AC = 22$ cm, $BM = 3$ cm, $DN = 3$ cm

Area of quadrilateral ABCD = Area of $\triangle ABC$ + Area of $\triangle ADC$

$$= \frac{1}{2} \times AC \times BM + \frac{1}{2} \times AC \times DN$$

$$= \frac{1}{2} \times 22 \times 3 + \frac{1}{2} \times 22 \times 3$$

$$= 3 \times 11 + 3 \times 11$$

$$= 33 + 33$$

$$= 66 \text{ cm}^2$$

Thus, the area of quadrilateral ABCD is 66 cm^2 .